

Traffic Behavior Profiling & Intelligence Report

Analysis Mode: Full

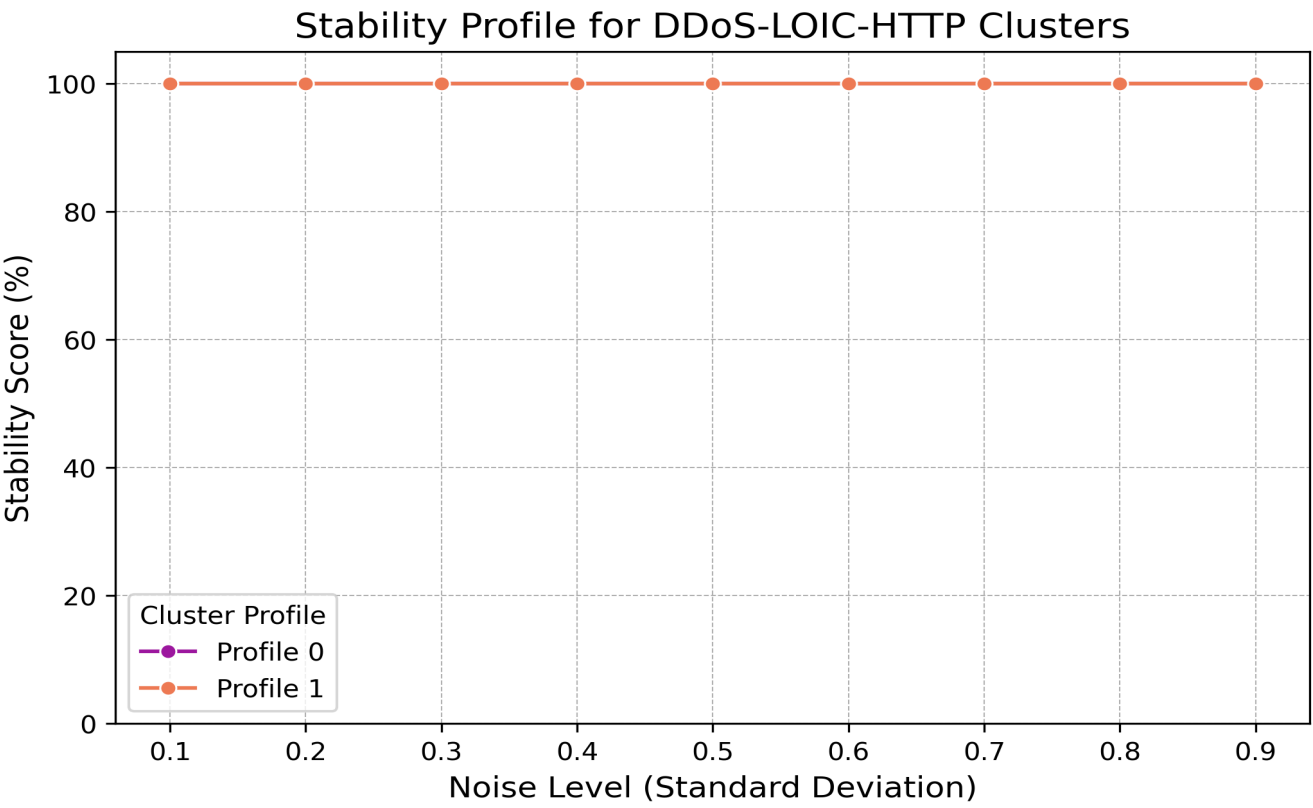
Executive Summary

This report details a scalable, multi-stage data analysis pipeline designed to transform raw network data into **actionable cybersecurity intelligence**. Leveraging a distributed computing framework (Apache Spark), the system processed 9167581 network records to automatically discover and profile distinct malicious and benign activities. The analysis pipeline consisted of: **(1) Data Ingestion and Preprocessing**, **(2) Feature Selection** using a tuned Decision Tree, **(3) Behavior Profiling** using a tuned K-Means clustering algorithm, **(4) a tuned Mitigation Recommendation Engine**, and **(5) Profile Stability Validation** via a Monte Carlo stress test. The key features for distinguishing traffic included **fwd_packet_length_max**, **init_fwd_win_bytes**, **total_backward_packets**. This project demonstrates a practical approach for Security Operations Centers (SOCs) to automate threat analysis at scale.

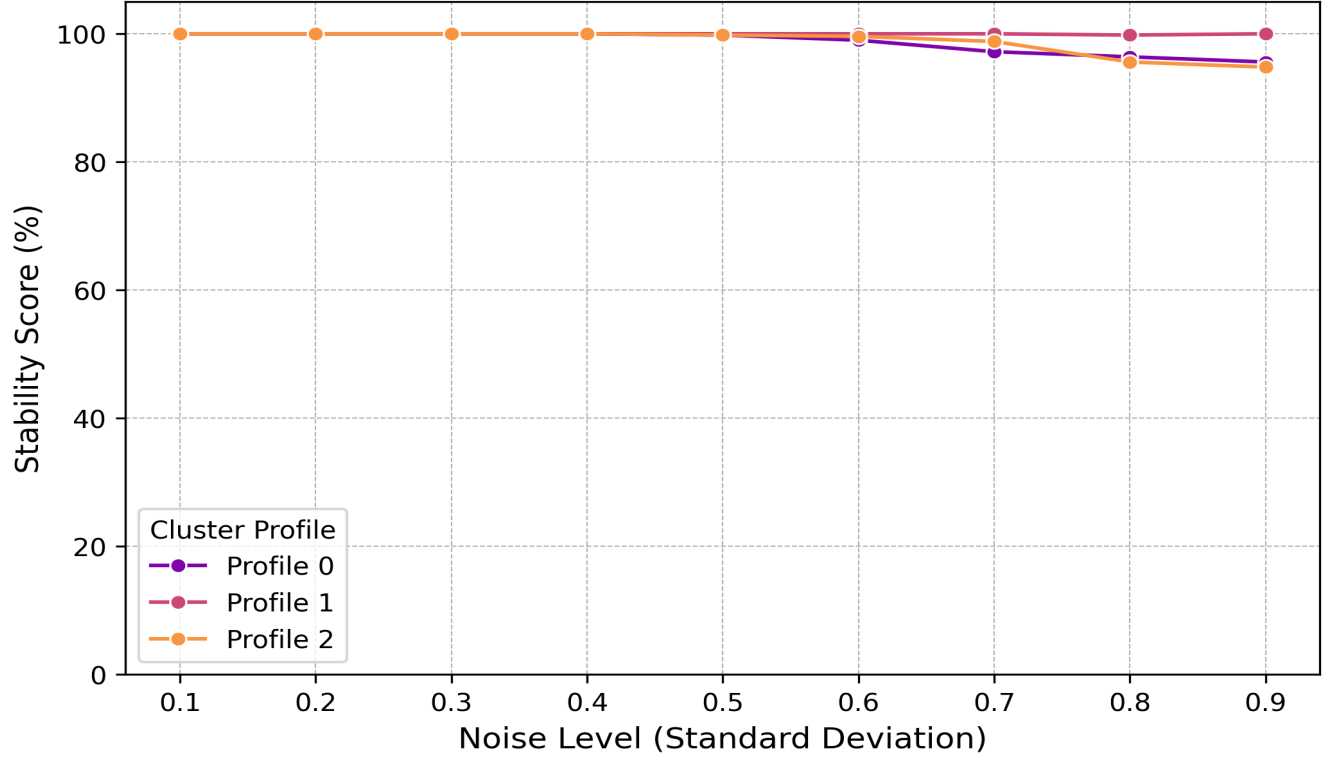
Automated Mitigation Recommendations

Threat Profile	Recommended Action 1	Recommended Action 2	Recommended Action 3
DDoS_P2	Rate-Limit-IPs	Blacklist-IP	Analyze-Payload
Botnet_P0	Rate-Limit-IPs	Blacklist-IP	Block-Port-80
Botnet_P1	Blacklist-IP	Analyze-Payload	Rate-Limit-IPs
DDoS_P3	Blacklist-IP	Rate-Limit-IPs	Analyze-Payload
DDoS_P4	Rate-Limit-IPs	Blacklist-IP	Analyze-Payload
DDoS-HOIC_P0	Rate-Limit-IPs	Blacklist-IP	Analyze-Payload
DoS-Hulk_P0	Rate-Limit-IPs	Blacklist-IP	Analyze-Payload
DDoS-HOIC_P1	Blacklist-IP	Rate-Limit-IPs	Analyze-Payload
DDoS-LOIC-HTTP_P0	Blacklist-IP	Rate-Limit-IPs	Analyze-Payload
DoS-Hulk_P1	Blacklist-IP	Rate-Limit-IPs	Analyze-Payload
DDoS-LOIC-HTTP_P1	Rate-Limit-IPs	Blacklist-IP	Analyze-Payload
DoS-Hulk_P2	Blacklist-IP	Rate-Limit-IPs	Analyze-Payload
DDoS-NTP_P0	Rate-Limit-IPs	Blacklist-IP	Analyze-Payload
DDoS-NTP_P1	Rate-Limit-IPs	Blacklist-IP	Analyze-Payload
DDoS_P0	Rate-Limit-IPs	Blacklist-IP	Analyze-Payload
DDoS_P1	Blacklist-IP	Rate-Limit-IPs	Analyze-Payload

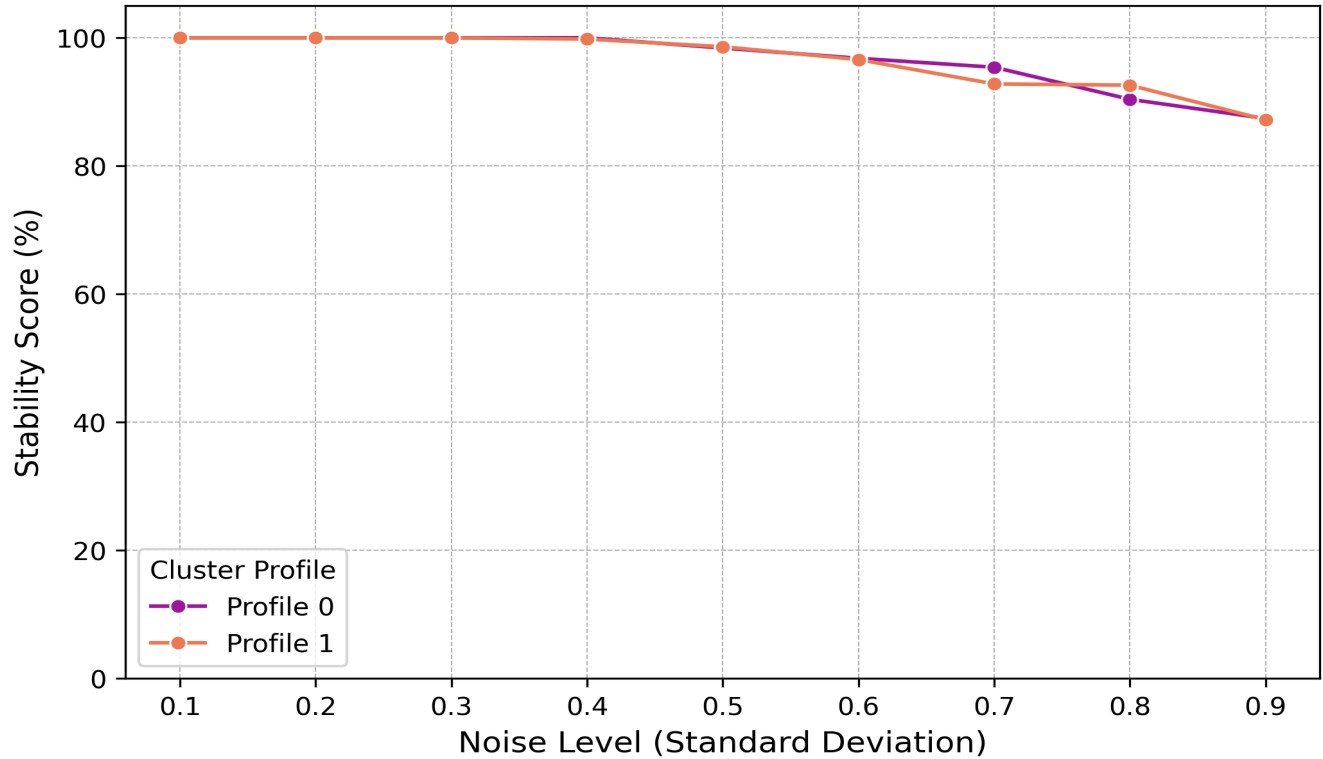
Profile Stability Stress Test (Monte Carlo)

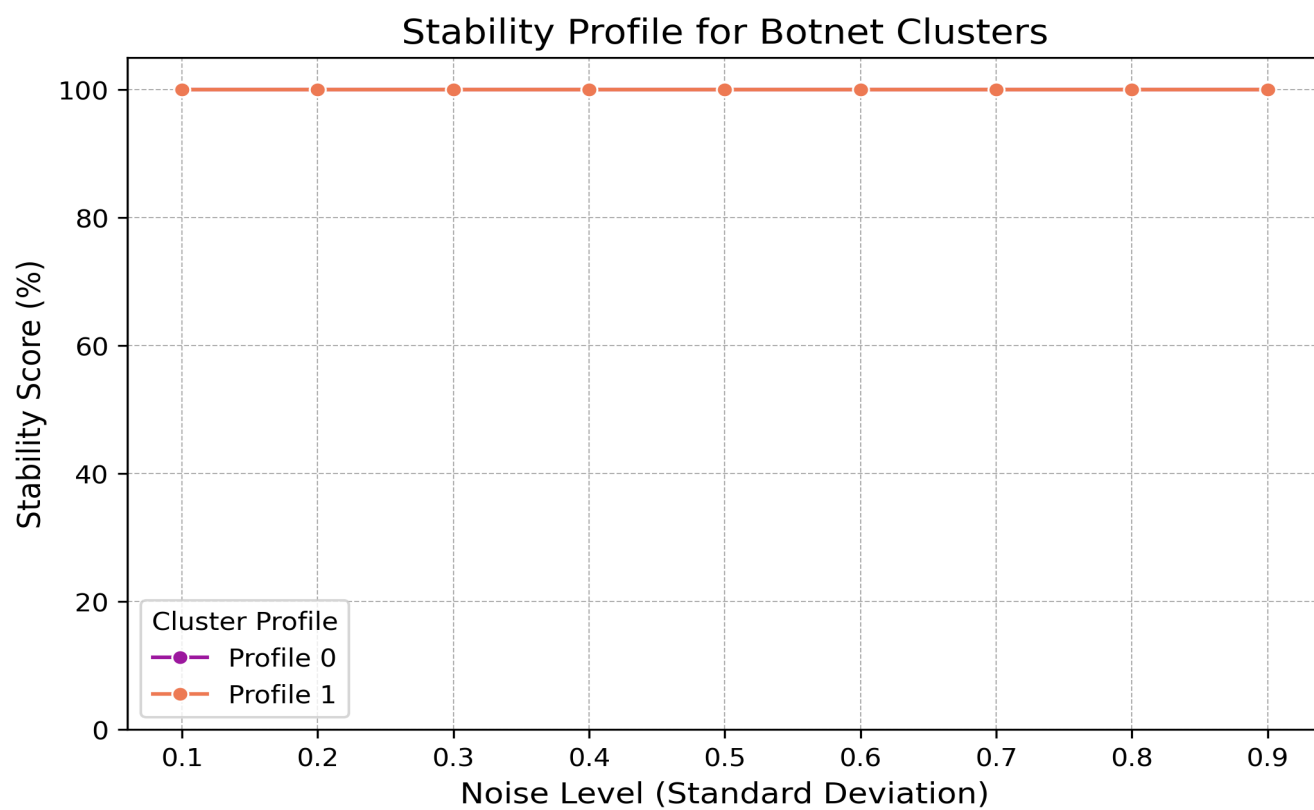


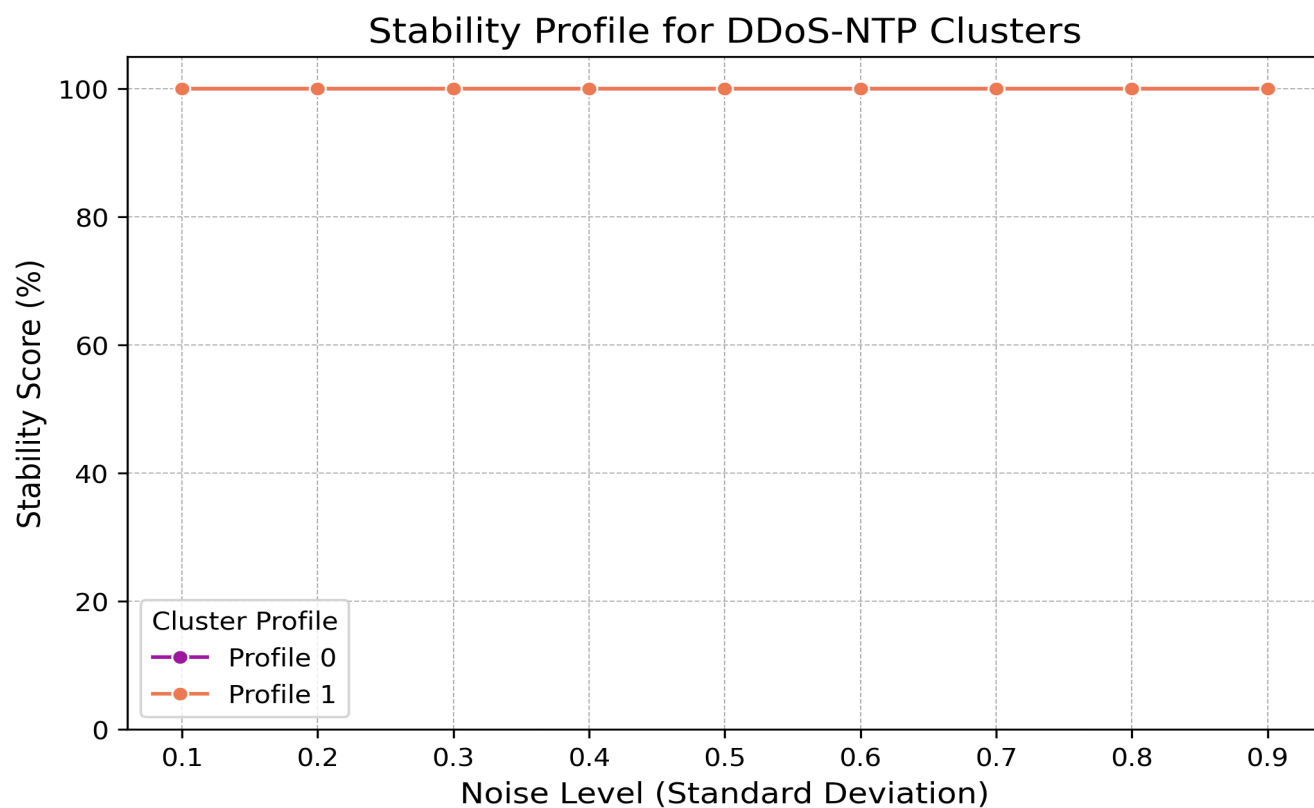
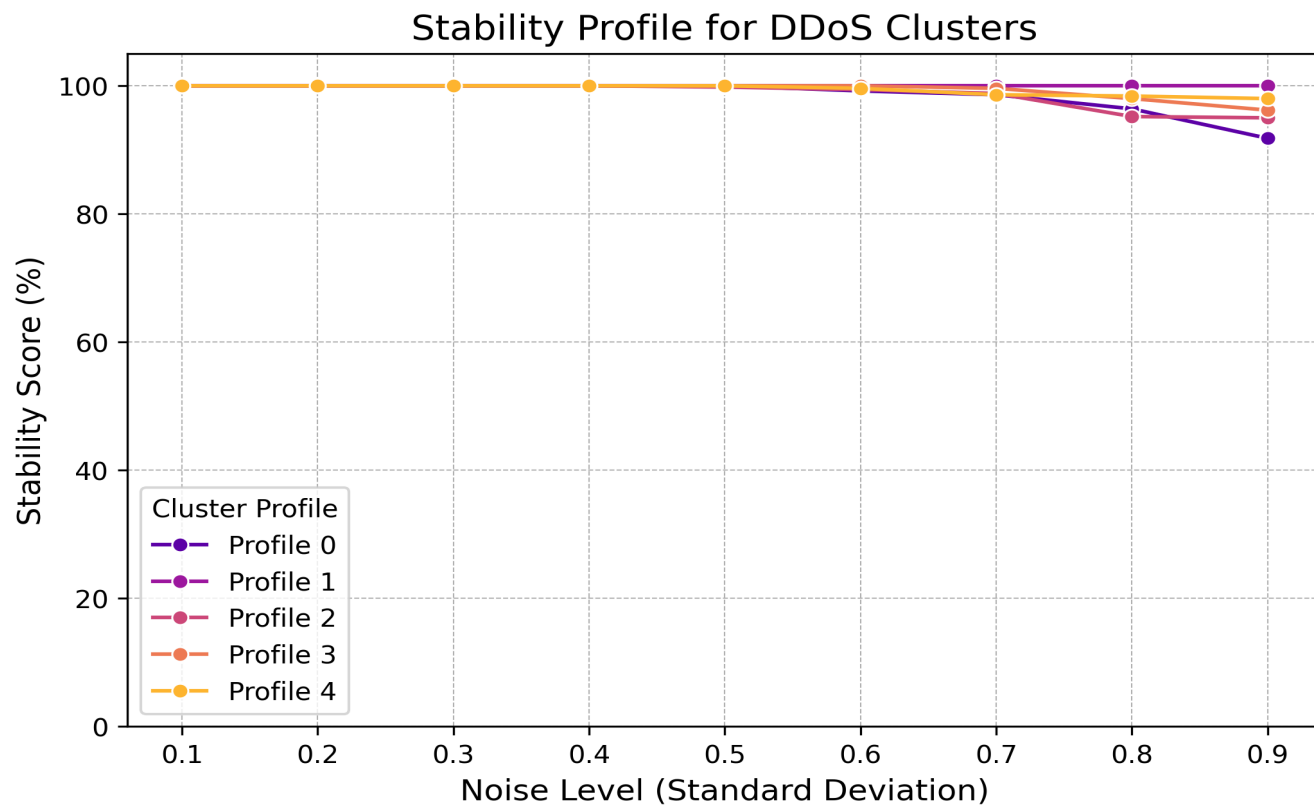
Stability Profile for DoS-Hulk Clusters



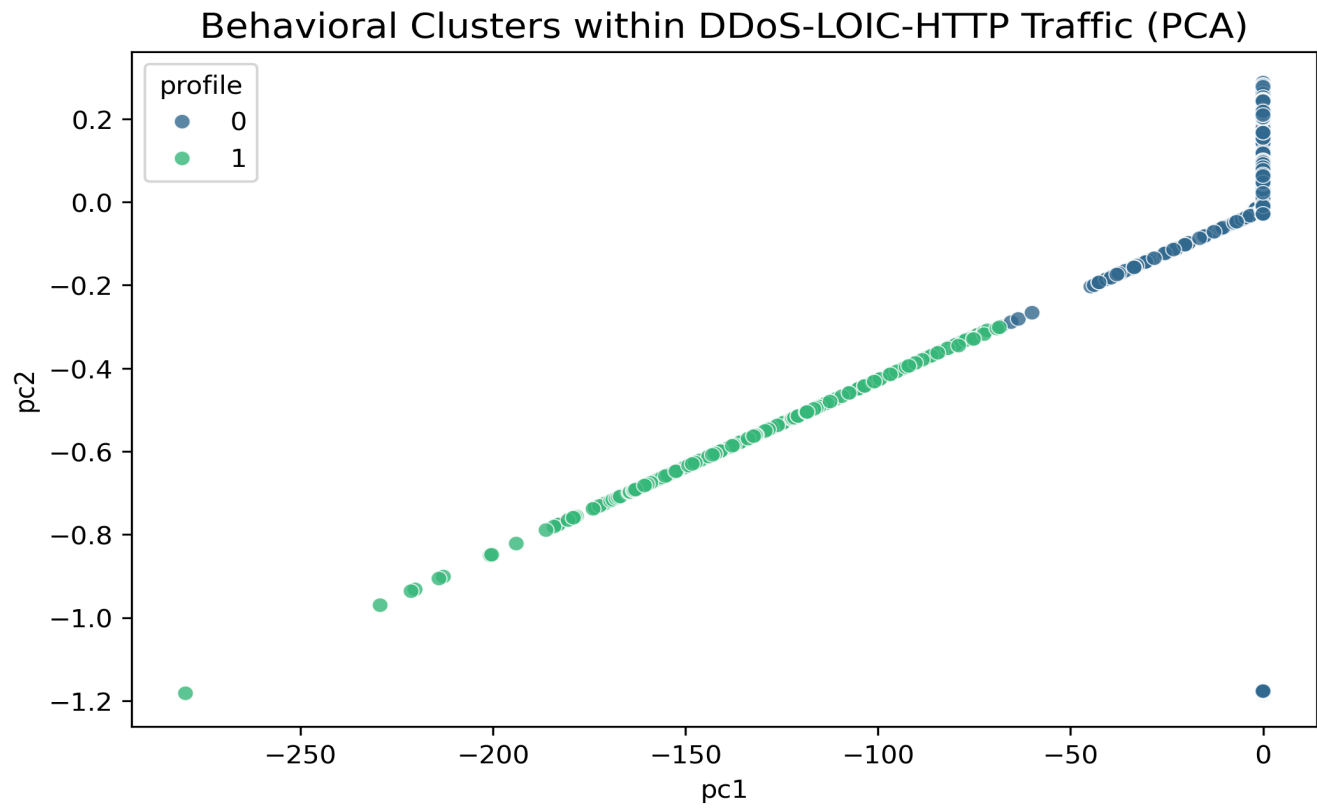
Stability Profile for DDoS-HOIC Clusters



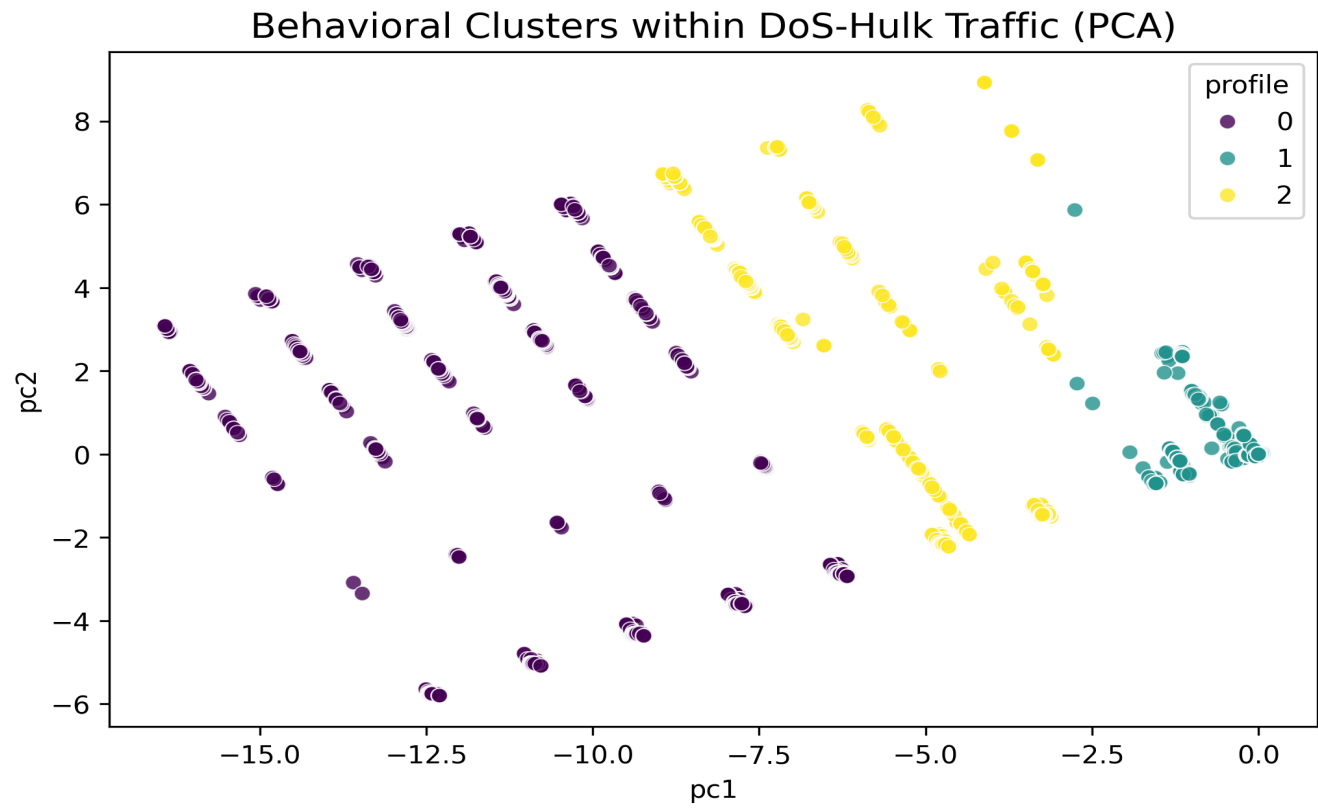




Detailed Profile: DDoS-LOIC-HTTP



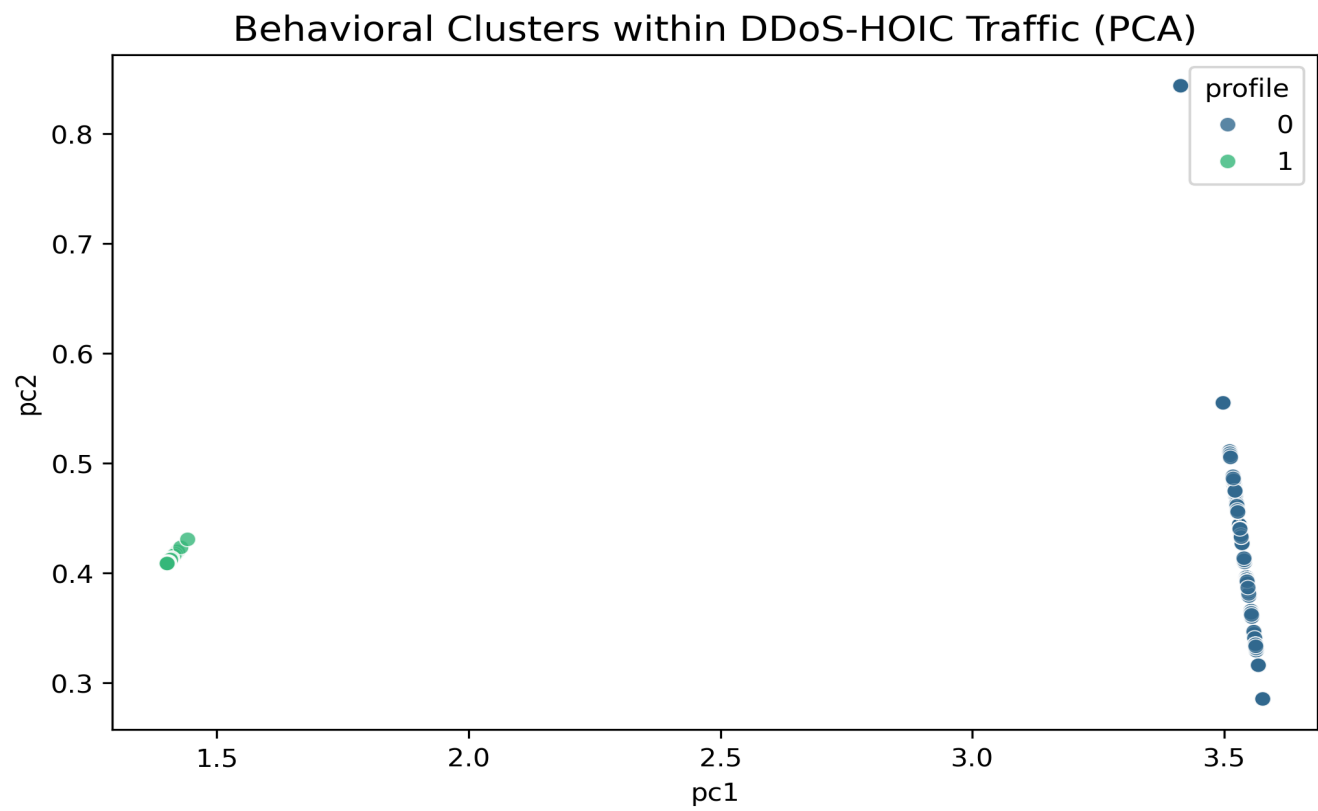
Detailed Profile: DoS-Hulk



Profile Fingerprints (Cluster Centers)

Feature	Profile 0	Profile 1	Profile 2
Fwd Packet Length Max	347.01	53.16	346.65
Init Fwd Win Bytes	2,842.34	2,563.17	2,839.26
Total Backward Packets	6.02	0.77	6.53
Fwd Packets Length Total	412.81	59.40	431.72
Bwd Packet Length Max	6,764.27	141.52	4,223.27
Fwd Header Length	210.62	89.87	211.30
Fwd Act Data Packets	1.86	0.25	1.89
Init Bwd Win Bytes	235.00	125.99	262.43
Fwd lat Min	150.22	23,287.43	253.59
Flow Duration	84,750,361.85	1,138,632.54	83,730,517.25

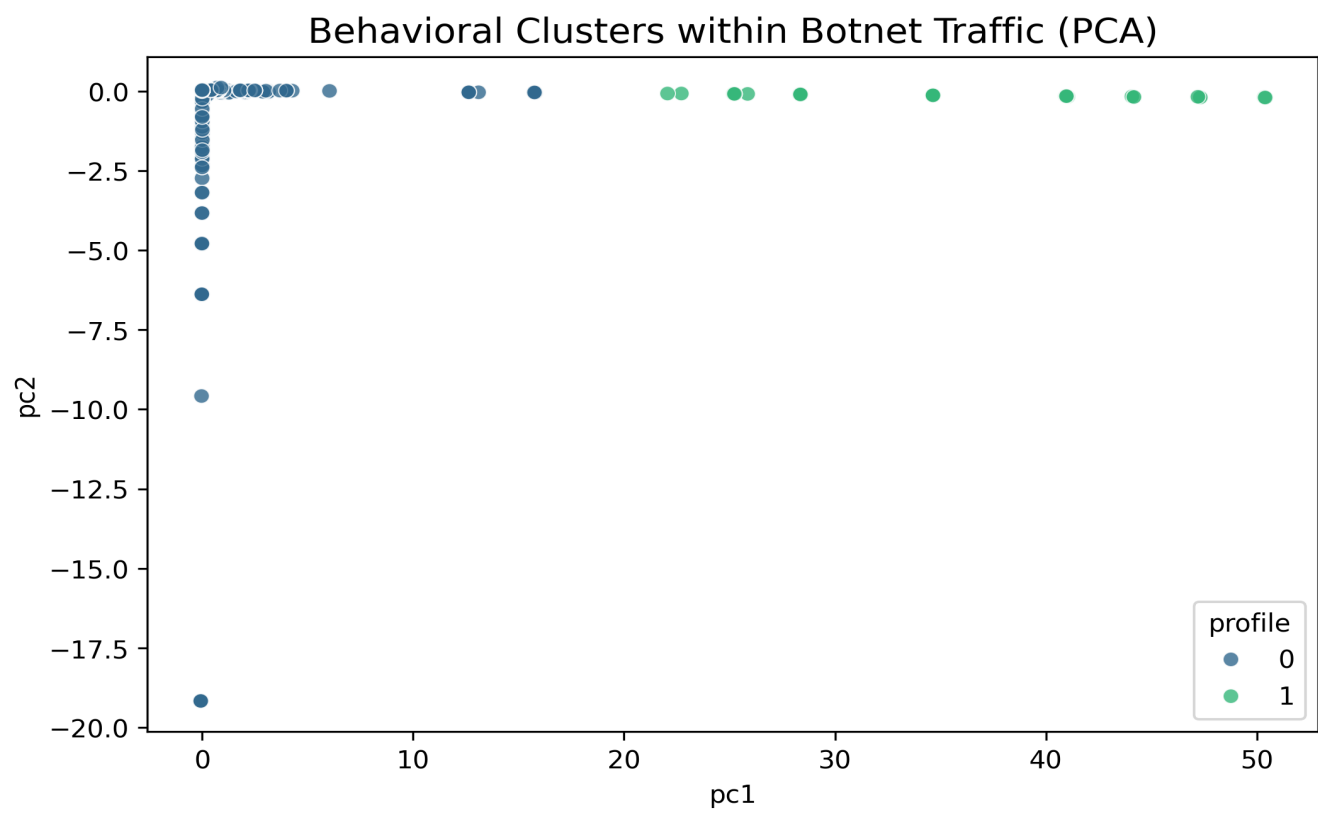
Detailed Profile: DDoS-HOIC



Profile Fingerprints (Cluster Centers)

Feature	Profile 0	Profile 1
Fwd Packet Length Max	299.12	0.00
Init Fwd Win Bytes	65,535.00	32,738.00
Total Backward Packets	4.00	0.00
Fwd Packets Length Total	299.12	0.00
Bwd Packet Length Max	935.00	0.00
Fwd Header Length	72.00	40.00
Fwd Act Data Packets	1.00	0.00
Init Bwd Win Bytes	218.99	-1.00
Fwd lat Min	321.88	18,815.72
Flow Duration	11,704.73	18,815.72

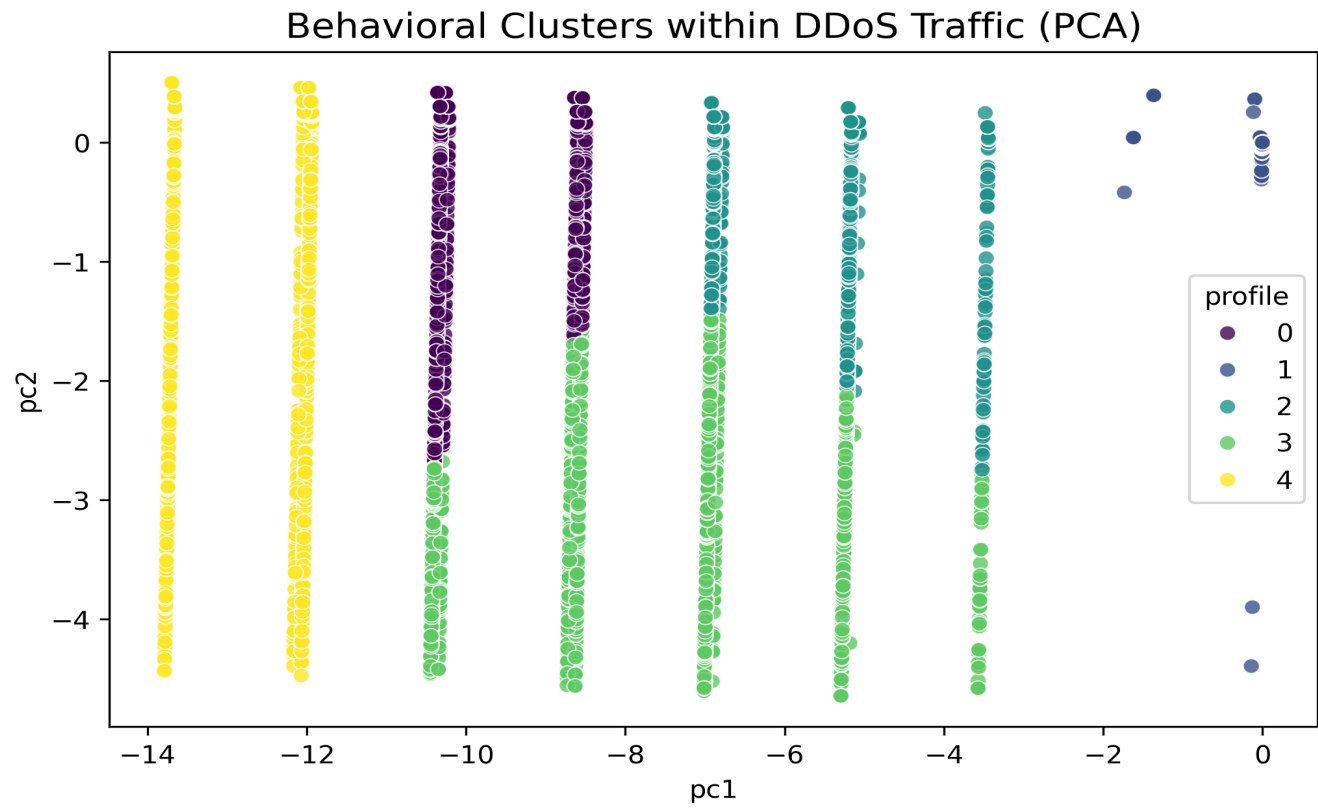
Detailed Profile: Botnet



Profile Fingerprints (Cluster Centers)

Feature	Profile 0	Profile 1
Fwd Packet Length Max	313.47	17,219.88
Init Fwd Win Bytes	8,108.06	8,192.00
Total Backward Packets	4.00	56.94
Fwd Packets Length Total	318.95	132,204.81
Bwd Packet Length Max	116.52	128.00
Fwd Header Length	73.24	598.88
Fwd Act Data Packets	1.02	28.34
Init Bwd Win Bytes	286.29	1,271.97
Fwd Iat Min	1,856.41	20.09
Flow Duration	151,856.30	116,433.47

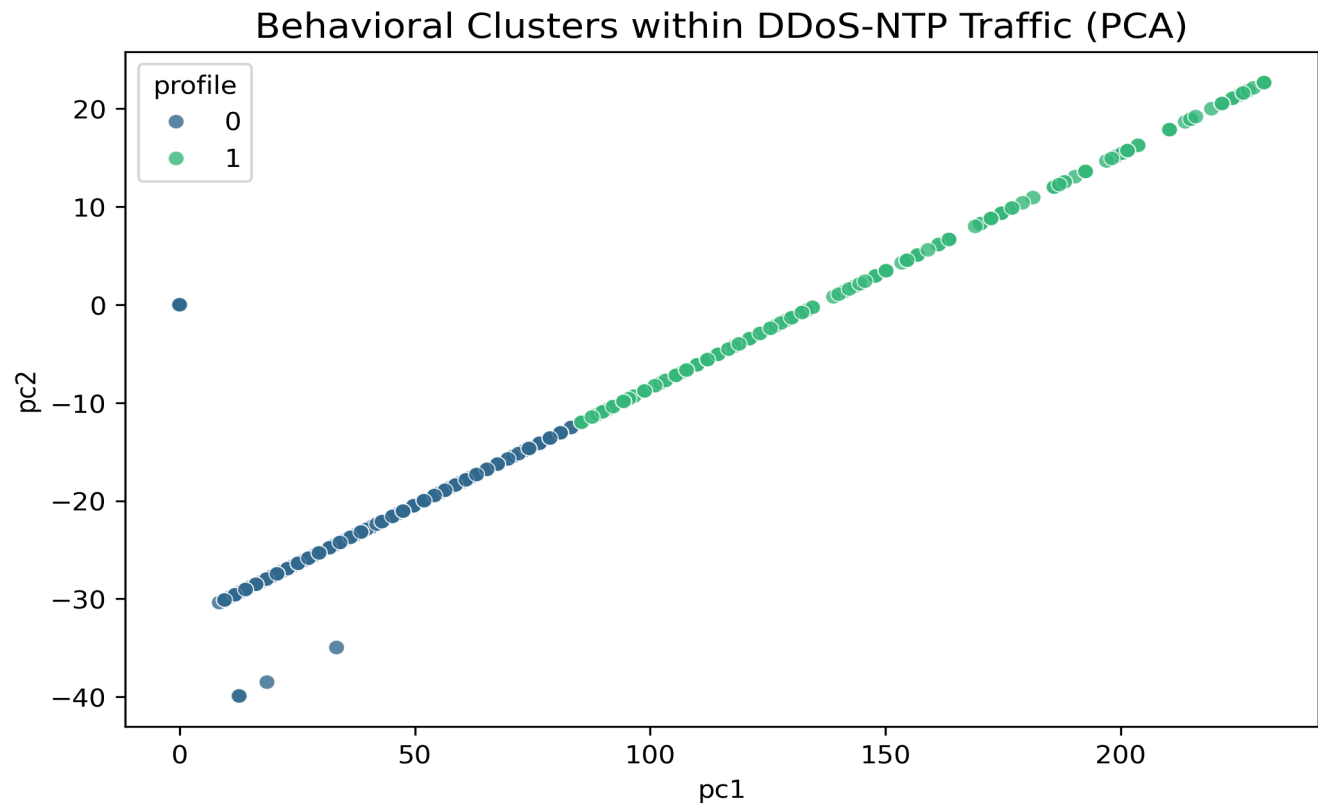
Detailed Profile: DDoS



Profile Fingerprints (Cluster Centers)

Feature	Profile 0	Profile 1	Profile 2	Profile 3	Profile 4
Fwd Packet Length Max	20.00	6.17	20.00	20.00	20.00
Init Fwd Win Bytes	6,358.66	282.24	6,451.17	256.00	6,031.16
Total Backward Packets	4.64	0.01	5.75	5.35	4.40
Fwd Packets Length Total	33.19	27.53	32.76	56.79	34.53
Bwd Packet Length Max	7,970.03	1.84	5,155.34	6,184.22	10,873.34
Fwd Header Length	96.05	90.43	94.60	174.67	100.54
Fwd Act Data Packets	2.96	3.52	2.90	6.13	3.14
Init Bwd Win Bytes	229.00	1.73	229.00	229.00	229.00
Fwd lat Min	378.13	6,662.68	460.10	27.68	131.57
Flow Duration	19,095,768.73	6,228,250.65	18,154,260.01	81,165,369.44	22,895,597.57

Detailed Profile: DDoS-NTP



Profile Fingerprints (Cluster Centers)

Feature	Profile 0	Profile 1
Fwd Packet Length Max	438.63	440.00
Init Fwd Win Bytes	52.93	-1.00
Total Backward Packets	0.02	0.00
Fwd Packets Length Total	29,570.54	69,033.80
Bwd Packet Length Max	0.80	0.00
Fwd Header Length	-145,711,272.35	-151,545,293,749.80
Fwd Act Data Packets	66.61	156.07
Init Bwd Win Bytes	5.73	-1.00
Fwd Iat Min	88.85	0.10
Flow Duration	15,573.43	13,430.94